

## Aim: Implementation of nesC Programming Model

Create a basic nesC application using modules and configurations to understand component wiring and interface binding.

### step 1: The Module (BlinkC.nc)

This component contains only the code logic. It does not know where the timer or LEDs come from; it just declares that it needs them.

#### CODE:

```
#include "Timer.h"

module BlinkC {
  // Interfaces used by this module
  uses interface Boot;
  uses interface Leds;
  uses interface Timer<TMilli> as Timer0;
}
implementation {
  event void Boot.booted() {
    // Command: start timer
    call Timer0.startPeriodic(500);
  }

  event void Timer0.fired() {
    // Command: toggle LED
    call Leds.led0Toggle();
  }
}
```

### Step 2: The Configuration / Wiring (BlinkAppC.nc)

This component has no logic. Its only job is component wiring (binding interfaces together).

Code:

```
configuration BlinkAppC {
}
implementation {
  // 1. Declare the components
  components MainC;
  components LedsC;
  components new TimerMilliC() as Timer0;
  components BlinkC as App;

  // 2. Wire the interfaces (Interface Binding)
  App.Boot -> MainC.Boot;
  App.Leds -> LedsC;
  App.Timer0 -> Timer0;
}
```

### Step 3: The Runner Script (sim.py)

#### CODE:

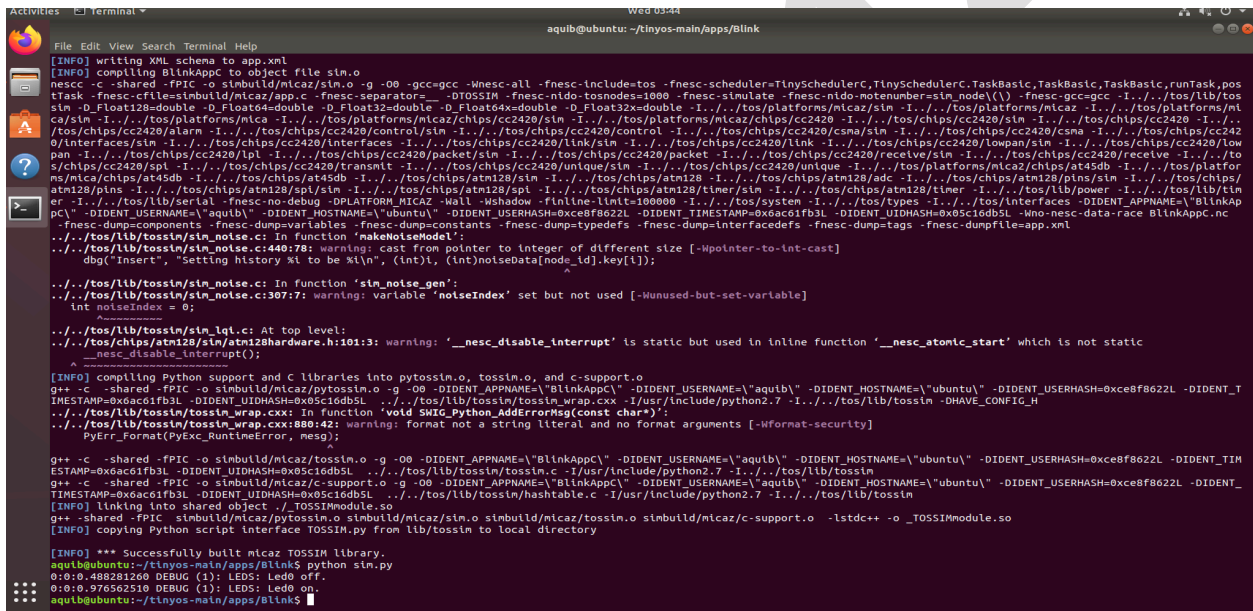
```
import sys
from TOSSIM import *

t = Tossim([])
t.addChannel("LedsC", sys.stdout)

# Boot mote 1
m = t.getNode(1)
m.bootAtTime(0)

# Run 20 events
for i in range(20):
    t.runNextEvent()
```

## OUTPUT:



```

[INFO] writing XML schema to app.xml
[INFO] compiling BlinkAppC to object file sin.o
nesc -c -shared -fPIC -o sinbuild/micaz/sin.o -g -O0 -gcc-gcc -Wnesc-all -fnesc-include-tos -fnesc-scheduler-TinySchedulerC,TinySchedulerC.TaskBasic,TaskBasic,runTask,pos
Task -fnesc-cfile=sinbuild/micaz/app.c -fnesc-separators= -DIOSSIM -fnesc-nido-tosnodes=1000 -fnesc-simulate -fnesc-nido-notenumber=sin node{(\)} -fnesc-gcc-gcc -I../tos/lib/tos
sin -D_Float128=double -D_Float64=double -D_Float32=double -D_Float64x=double -D_Float32x=double -I../tos/platforms/micaz/sin -I../tos/platforms/micaz -I../tos/platforms/mi
ca/sin -I../tos/platforms/mica -I../tos/platforms/micaz/chips/cc2420/sin -I../tos/platforms/micaz/chips/cc2420 -I../tos/chips/cc2420/sin -I../tos/chips/cc2420 -I../
tos/chips/cc2420/alarm -I../tos/chips/cc2420/control/sin -I../tos/chips/cc2420/control -I../tos/chips/cc2420/csm/sin -I../tos/chips/cc2420/csm -I../tos/chips/cc242
0/interfaces/sin -I../tos/chips/cc2420/interfaces -I../tos/chips/cc2420/link/sin -I../tos/chips/cc2420/link -I../tos/chips/cc2420/lowpan/sin -I../tos/chips/cc2420/low
pan -I../tos/chips/cc2420/lpl -I../tos/chips/cc2420/package/sin -I../tos/chips/cc2420/package -I../tos/chips/cc2420/receive/sin -I../tos/chips/cc2420/receive -I../to
s/chips/cc2420/spt -I../tos/chips/cc2420/transmit -I../tos/chips/cc2420/unique/sin -I../tos/chips/cc2420/unique -I../tos/platforms/micaz/chips/at45db -I../tos/platfor
ms/micaz/chips/at45db -I../tos/chips/at45db -I../tos/chips/atm128/sin -I../tos/chips/atm128 -I../tos/chips/atm128/adc -I../tos/chips/atm128/pins/sin -I../tos/chips/
atm128/pins -I../tos/chips/atm128/spl/sin -I../tos/chips/atm128/spl -I../tos/chips/atm128/timer/sin -I../tos/chips/atm128/timer -I../tos/lib/power -I../tos/lib/tim
er -I../tos/lib/serial -fnesc-no-debug -DPLATFORM_MICAZ -Wall -Wshadow -finline-limit=100000 -I../tos/system -I../tos/types -I../tos/interfaces -DIDENT_APPNAME="BlinkAp
pC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DIDENT_TIMESTAMP=0x6ac01fb3L -DIDENT_UIDHASH=0x05c16db5L -Wno-nesc-data-race BlinkAppC.nc
-fnesc-dump-components -fnesc-dump-variables -fnesc-dump-constants -fnesc-dump-typedefs -fnesc-dump-interface-defs -fnesc-dump-tags -fnesc-dump-file=app.xml
../tos/lib/tossin/sin_noise.c:440:78: warning: cast from pointer to integer of different size [-Wpointer-to-int-cast]
    dbg("Insert", "Setting history %i to be %i\n", (int)i, (int)noiseData[node_id].key[i]);
    ^
../tos/lib/tossin/sin_noise.c: In function 'sin_noise_gen':
../tos/lib/tossin/sin_noise.c:307:7: warning: variable 'noiseIndex' set but not used [-Wunused-but-set-variable]
    int noiseIndex = 0;
    ^
../tos/lib/tossin/sin_lqi.c: At top level:
../tos/chips/atm128/sin/atm128hardware.h:101:3: warning: '__nesc_disable_interrupt' is static but used in inline function '__nesc_atomic_start' which is not static
    __nesc_disable_interrupt();
    ^
[INFO] compiling Python support and C libraries into pytosin.o, tossin.o, and c-support.o
g++ -c -shared -fPIC -o sinbuild/micaz/pytosin.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DIDENT_T
IMESTAMP=0x6ac01fb3L -DIDENT_UIDHASH=0x05c16db5L -I../tos/lib/tossin/tossin_wrap.cxx -I/usr/include/python2.7 -I../tos/lib/tossin -DHAVE_CONFIG_H
../tos/lib/tossin/tossin_wrap.cxx: In function 'void SWIG_Python_AddErrorMsg(const char*)':
../tos/lib/tossin/tossin_wrap.cxx:880:42: warning: format not a string literal and no format arguments [-Wformat-security]
    PyErr_Format(PyExc_RuntimeError, msg);
                                         ^
g++ -c -shared -fPIC -o sinbuild/micaz/pytosin.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DIDENT_T
IMESTAMP=0x6ac01fb3L -DIDENT_UIDHASH=0x05c16db5L -I../tos/lib/tossin/tossin.c -I/usr/include/python2.7 -I../tos/lib/tossin
g++ -c -shared -fPIC -o sinbuild/micaz/c-support.o -g -O0 -DIDENT_APPNAME="BlinkAppC" -DIDENT_USERNAME="aquib" -DIDENT_HOSTNAME="ubuntu" -DIDENT_USERHASH=0xce8f8622L -DIDENT_
TIMESTAMP=0x6ac01fb3L -DIDENT_UIDHASH=0x05c16db5L -I../tos/lib/tossin/hashtable.c -I/usr/include/python2.7 -I../tos/lib/tossin
[INFO] linking into shared object ./TOSSIMmodule.so
g++ -shared -fPIC -o sinbuild/micaz/pytosin.o sinbuild/micaz/sin.o sinbuild/micaz/pytosin.o sinbuild/micaz/c-support.o -lstdc++ -o TOSSIMmodule.so
[INFO] copying Python script interface TOSSIM.py from lib/tossin to local directory
[INFO] *** Successfully built micaz TOSSIM library.
aquib@ubuntu:~/tinyos-main/apps/Blink$ python sin.py
0:0:0.488281260 DEBUG (1): LEDS: Led0 off.
0:0:0.976562510 DEBUG (1): LEDS: Led0 on.
aquib@ubuntu:~/tinyos-main/apps/Blink$
```